Srinivas Aditya Abbaraju

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.71
Sri Chaitanya Junior Kalasala	Class XII (Telangana State Board of Intermediate Education)	2020	98.70%
Sri Chaitanya School	Class X (Board of Secondary Education, Telangana)	2018	10.00%

SKILLS

- **Programming:** Python (proficient), C++, Java (basic), JavaScript (basic), HTML, CSS
- **Frameworks & Libraries:** PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, ChromaDB
- **Web Technologies:** React.js, Flask, FastAPI
- **Tools:** Power BI, Git, VS Code, Jupyter, Linux/Bash, Ollama, Streamlit
- **ML Areas:** Computer Vision, DL, Domain Adaptation, Hyperspectral Imaging, LLM's, VLMs, GenAI, RAG, LLM Agents

PUBLICATIONS

1. *"Domain Adaptation In Agricultural Hyperspectral Imaging: Advancing Robust Cross-Domain Learning For Precision Agriculture",* IGARSS 2026
2. *"Comparative Evaluation of Band Selection Methods For Early Crop Stress Detection Using UAV-Based Hyperspectral Imaging",* IGARSS 2026
3. *"A Survey of Band Selection Techniques for Hyperspectral Crop Stress Detection"*, InGARSS 2026
4. *"SIGMA-Based RGB-Hyperspectral Fusion for Semantic Segmentation in Autonomous Driving",* ICIP 2026
5. *"A Review on Lane Detection for Autonomous Vehicles",* FICTA, Springer, DOI: 10.1007/978-981-96-0143-1_19
6. *"Cyberbullying Detection Using Machine Learning and Deep Learning Models"* , Authorea, DOI: 10.22541/au.170664263.38254624/v1

THESIS

Cross-Crop Generalization for Early Stress Detection via Hyperspectral Imaging

- Designing a Mamba-based hyperspectral foundation model (single- and dual-encoder variants) across 7 stress/classification tasks spanning 3 crops (groundnut, pearl millet, sorghum); achieved accuracy competitive with or exceeding HyperSigma, SpectralEarth, and SpectralGPT using substantially fewer parameters
- Developing a prototype- and disentanglement-based unsupervised domain adaptation framework to generalize early stress detection across crops
- Preprocessed and curated raw UAV HSI data (PIKA Resonon-L, 400-1000nm) from BIL to NumPy format for 3 ICRISAT crop datasets (2021-22 collection cycle)
- Identified common discriminative spectral sub-bands (460-560nm, 600-650nm, 700-790nm) generalizable across multiple crops through comparative band-selection experiments, informing model compression and sensor design choices
- Contributing to an ongoing HSI data generation effort aimed at reducing dependency on expensive, time-intensive hyperspectral data acquisition

PROJECTS

Real-Time Lane Mark Detection on Edge-AI, Autonomous Ground Vehicle

- Deployed and benchmarked 3 lightweight lane-detection architectures (ENet, YOLOP, TwinLiteNet+) in real time on an NVIDIA Jetson Orin, achieving up to 93.5% ± 4.5% detection confidence and 30 FPS on IITH's TIHAN

autonomous vehicle testbed and public national highways
- Evaluated models across 5 public benchmarks (BDD100K, KITTI, Mapillary Vistas, TuSimple, CurveLane) using precision, recall, F1, IoU, and FPS to characterize accuracy-latency trade-offs for deployment
- Co-annotated ~3,000 images (1,500 individually) for current-lane/alternate-lane road segmentation using CVAT, supporting a PSPNet model trained on BDD100K for road segmentation which helps in autonomous navigation and obstacle avoidance
- Derived lane-center deviation using B-splines and convex hull geometry

Lane Detection Using Classical and Deep Learning Methods

- Implemented and benchmarked 3 ego-lane detection approaches: classical CV using perspective transformation, a lightweight encoder-decoder model, and ENet
- Integrated a pretrained YOLOv8 object detection model and derived the ego-vehicle's deviation from the road center and heading direction using geometric/mathematical formulations
- Built a Flask-based web interface enabling users to upload driving footage and automatically download lane-detection outputs

Diffusion-Driven Data Replay for Federated Class-Continual Learning

- Reproduced the DDDR (ECCV 2024) framework from scratch for federated class-continual learning, matching published accuracy and forgetting metrics within standard deviation across CIFAR-100 and TinyImageNet, 3 random seeds
- Designed a novel VAE-based privacy-preserving replay mechanism that obfuscates synthetic replay images while retaining class-discriminative semantics, limiting accuracy loss to 3-6% under stronger privacy constraints
- Ran an ablation across loss components (Knowledge Distillation, Past Cross-Entropy, Supervised Contrastive Loss) and backbone variants (ResNet18/34/50-CBAM), identifying KD as most critical to preventing catastrophic forgetting
- Evaluated classifier-level differential privacy noise injection, showing it degrades performance sharply at high privacy budgets compared to the proposed replay-level approach

Research Paper Assistant: Grounded RAG with LLM Agent

- Built a local RAG assistant using hybrid retrieval (BGE embeddings + BM25) that cites paper and page, and labels any answer not from the library as general knowledge.
- Designed an LLM tool-routing agent (library Q&A, arXiv paper search, literature review, library overview) with 96% routing accuracy (24/25). Downloads need user confirmation. Code check confirms the agent cannot download on its own.
- Evaluated on 22 hand-verified questions: 59% acceptable answers (5% hallucination) with Llama 3.1 70B, 50% with 8B.
- Added a weekly new-papers digest per research folder, with incremental indexing that re-embeds only changed PDFs.

Stellar Classification from Spectral Data

- Built an ANN classifier to distinguish stars, galaxies, and quasars from SDSS spectral data (~100K records, 18 raw features), narrowing to 8 informative features via feature selection and outlier removal
- Addressed class imbalance across 3 dataset scenarios, unbalanced, SMOTE-balanced, and randomly under-sampled, training and evaluating separately on each
- Ran a systematic hyperparameter search (learning rate, batch size, optimizer, activation function, hidden-layer architecture) across 288 configurations per scenario (864 models total) via stratified 5-fold cross-validation
- Achieved best validation accuracy of 97.46% (SMOTE-balanced) with per-class ROC-AUC of 0.99-1.00, identifying SGD as a consistently underperforming optimizer

Sensor-Based Human Activity Classification

- Built and compared MLP, RNN, LSTM classifiers on 4-second PIR sensor time-series data for human activity classification
- Conducted EDA to identify >75% correlation-based redundancy among adjacent sensor features; applied alternating feature selection and 3-feature averaging as engineering strategies to reduce dimensionality
- Compared performance of both engineered feature sets against the original dataset, finding negligible accuracy difference while reducing computational cost
- Implemented stratified 5-fold cross-validation with a reproducible training/evaluation pipeline, evaluating models via accuracy, macro-F1 score, and confusion-matrix analysis, with learning curves visualized for diagnostics

Cyberbullying Detection and Classification

- Built a text classification pipeline to detect cyberbullying and categorize it into 5 types (religion, racism, gender, caste, other) from social media comments
- Cleaned and tokenized raw text data using standard NLP preprocessing (stopword removal, normalization, vectorization) before model training
- Implemented and compared classical ML models (Random Forest, SVM) against deep learning (CNN, RNN) and transformer-based (BERT) approaches to benchmark performance across model families
- Published the comparative study as a peer-reviewed paper (Authorea), DOI: 10.22541/au.170664263.38254624/v1

Resume Builder

- Architected and built a full-stack resume builder using React.js (frontend) and Node.js (backend), with a dynamic form system for structured user input
- Implemented template-based resume generation supporting multiple layout formats through reusable UI components
- Designed real-time preview functionality, allowing users to see formatted resume output update live as they edit form

SCHOLASTIC ACHIEVEMENTS

- Gold Medal, B.Tech in CSE (AI/ML specialization) - top of department
- Winner, Invesco Placement Hackathon
- Developed backend logic for financial metric computation using real-time data fetched via Yahoo Finance API
- Implemented server-side processing to compute company-specific financial indicators based on given formulas
- Collaborated with frontend developers to integrate backend APIs into a complete end-to-end system

CERTIFICATIONS

- Data Analyst Intern Certificate: Bluestock Fintech (May 2026 - Jul 2026)
- NPTEL: Programming in Modern C++ (Silver Medal)
- NPTEL: Data Mining (Silver Medal)
- Coursera: Machine Learning Specialization
- Coursera: Google Data Analytics
- Coursera: Mathematics for Machine Learning Specialization
- Participation: Convergence 2k21/2k22 coding contest (VNRVJIET), Blackhole Simulation Hackathon

RELEVANT COURSES

Mathematics Courses: Matrix Theory and Linear Algebra, Probability and Random Variables, Stochastic Processes, Convex Optimization

Artificial Intelligence & Machine Learning Courses: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Science Analysis

Computer Science Courses: Data Structures and Algorithms, Advanced Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, Compiler Design, Computer Organization, Formal Language on Automata Theory

EXTRACURRICULAR ACTIVITIES

- Playing Video games and watching anime
- Cook and experiment with recipes, from minor variations to new dishes
- Interested in reading blogs and articles related to medicine and astronomy
- Design, build, and deploy personal full-stack web applications and games (Vercel, Render) as a hobby
- Provide informal event coordination and logistics support for friends' events